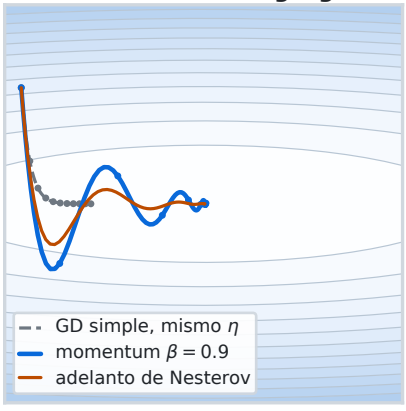
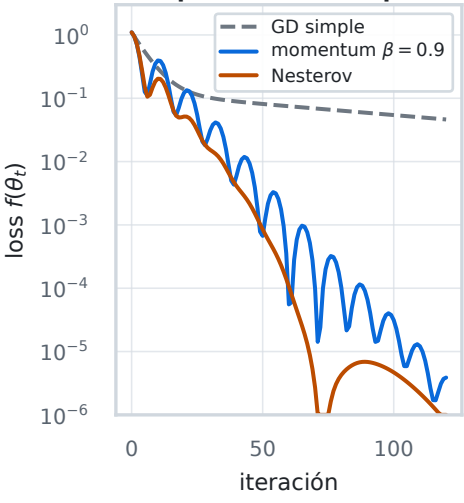
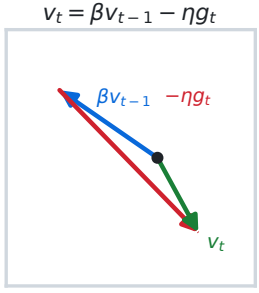

(a) se filtra el zigzag
GD simple, mismo η
momentum $\beta = 0.9$
adelanto de Nesterov


(b) mismo paso: velocidad por el fondo
GD simple
momentum $\beta = 0.9$
Nesterov
loss $f(\theta_t)$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
50
100
iteración


$v_t = \beta v_{t-1} - \eta g_t$
βv_{t-1}
$-\eta g_t$
v_t